EMPLOYEE ATTRITION PREDICTION SYSTEM

PRESENTED BY: RITHVIK KANOORI
ROLL NUMBER: 2403A51197
COURSE: MACHINE LEARNING
FACULTY: SOUTRICK DAS

PROBLEM STATEMENT

1. High employee attrition is a problem
2. Causes financial loss
3. **Manual prediction is difficult and inaccurate**
4. Affects productivity

PROJECT OBJECTIVE

1.To predict employee attrition using Machine Learning

2.To identify employees who are likely to leave the company

3.To help organizations improve employee retention

4.To compare different Machine Learning classification algorithms

FEATURES USED

1. Age
2. Income
3. Job satisfaction and years of experience
4. Overtime

ALGORITHMS SELECTED

LOGISTIC REGRESSION

predicts probability of attrition.

DECISION TREE

classifies using decision rules

RANDOM FOREST

Improves accuracy using multiple trees

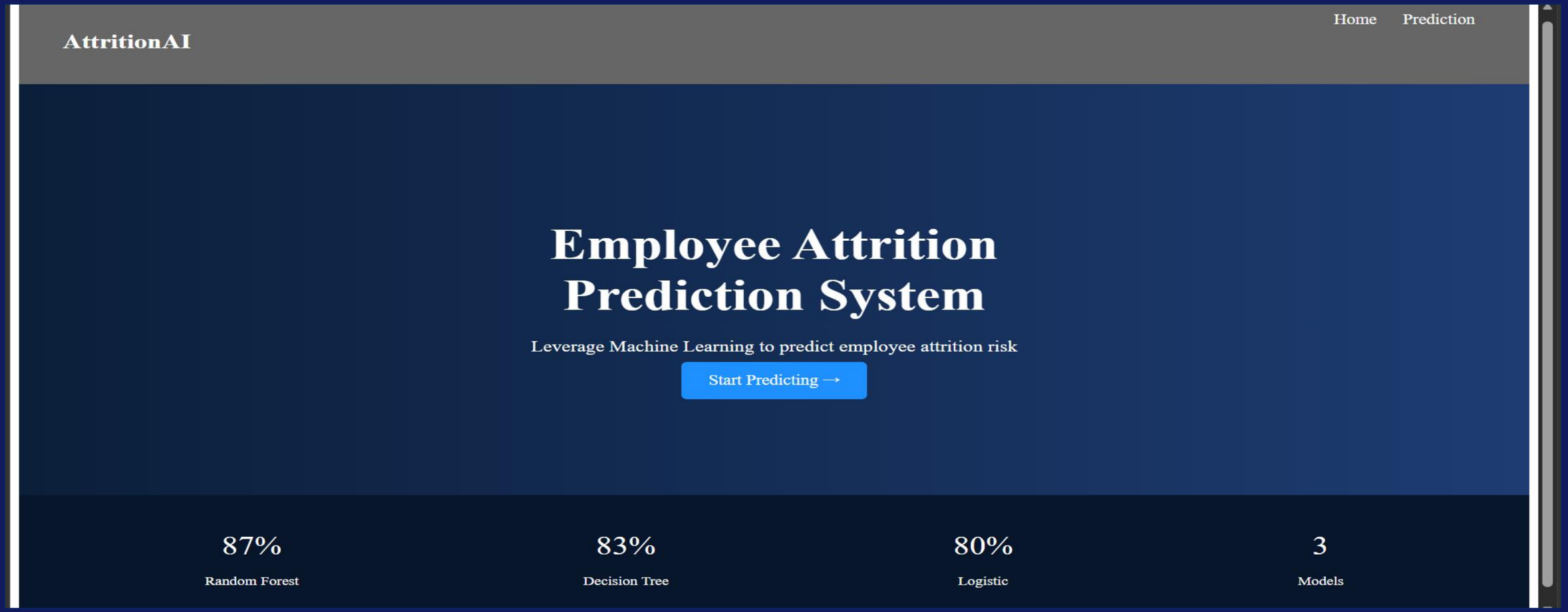
TECH STACK USED

Python
Flask
Scikit-learn
Pandas
HTML/CSS

WORKING

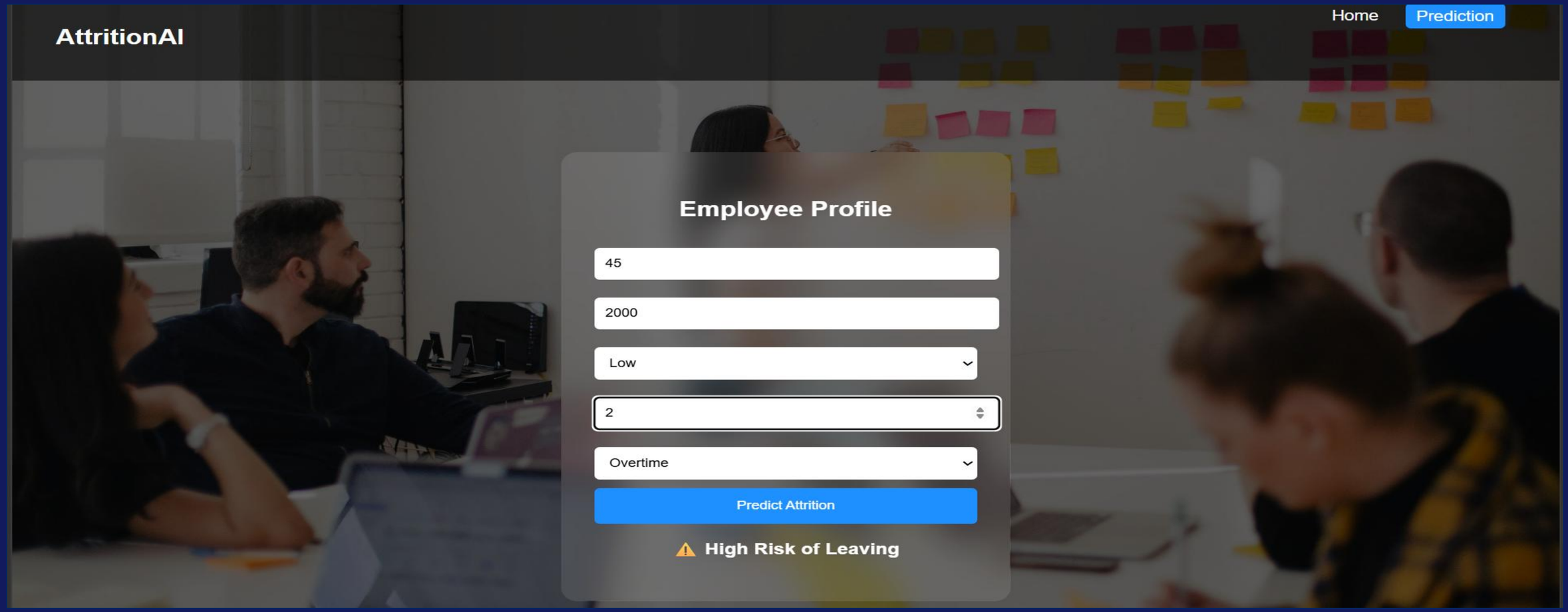
Input → Processing → ML Model → Output

USER INTERFACE



[Home Page Screenshot]

OUTPUT



Prediction Page Screenshot

Model Results & Predictions

The system predicts employee attrition based on input features.

Sample Input:
- Age: 30
- Monthly Income: 40000
- Job Satisfaction: High
- Years at Company: 5
- Overtime: Yes
Prediction Output:
⚠️ **High Risk of Leaving**
The model analyzes patterns and provides instant predictions through the web interface.

Model Performance & Accuracy

Algorithm Used:
Random Forest Classifier

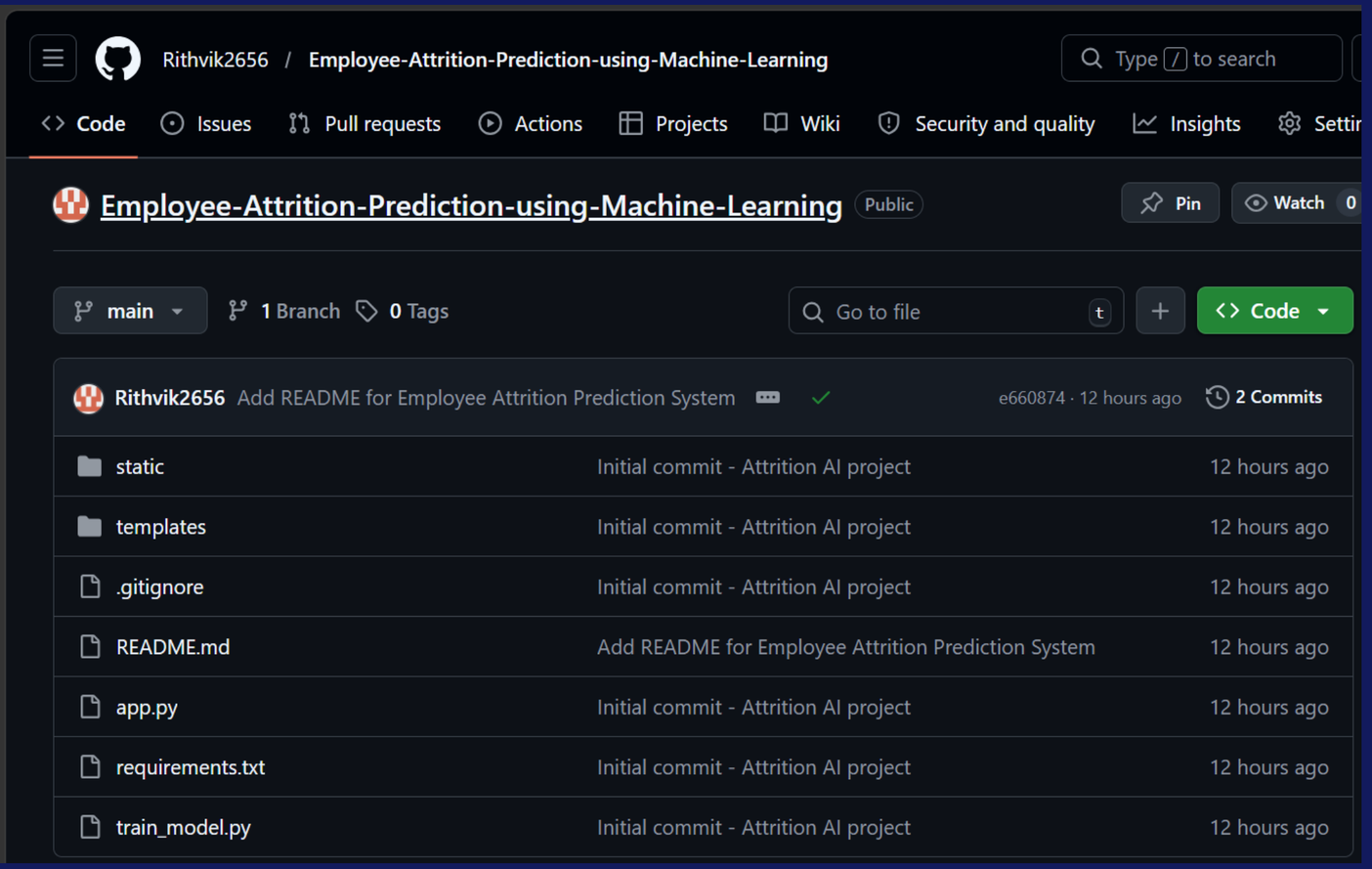
Model Accuracy:
~85% (approximate performance)

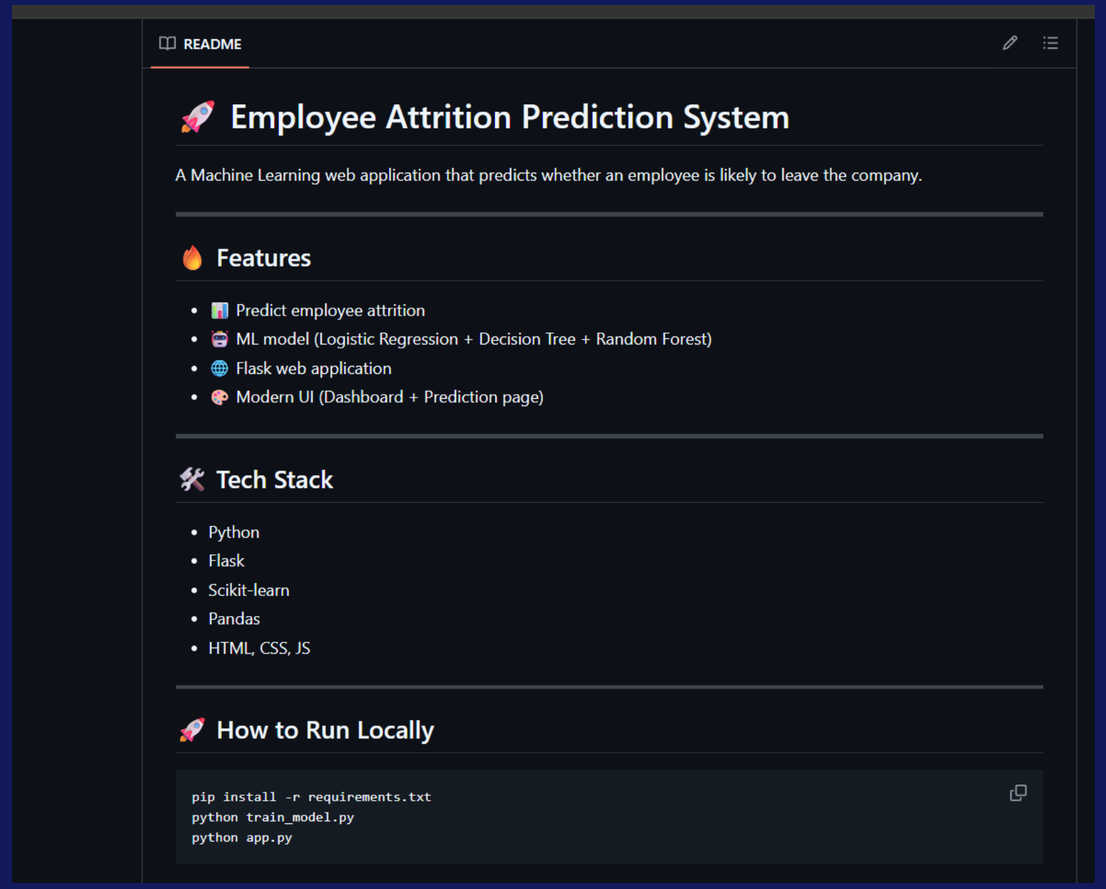
Other Models Tested:
- Logistic Regression: ~78%
- Decision Tree: ~80%

Conclusion:
Random Forest performs best with higher accuracy and stability.

GITHUB

https://github.com/Rithvik2656/Employee-Attrition-Prediction-using-Machine-Learning





THANK YOU